

Karim El Yaagoubi

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PROFILE

Software and Data Engineer with experience in developing applications, managing data systems, and building efficient technology solutions. Passionate about creating reliable, scalable products.

PROFESSIONAL EXPERIENCE

Astound Technology, *Backend Software Engineer intern* Jul 2026 – Oct 2026 | Casablanca

- Developed RESTful API using **Spring Boot** following layered architecture and clean coding practices.
- Designed and implemented data models with **MongoDB** and **PostgreSQL**.
- Integrated authentication and authorization using **Keycloak** with JWT and role based access control.
- Integrated API with **Next.js** applications.
- Used Git for version control and participated in code reviews.

PROJECTS

Semantic Modeling & Analytics of Cancer Data | Python, RDF, SPARQL, OWL

Feb 2026 – Present

Academic project

- Designed a custom OWL ontology modeling breast cancer knowledge across 5 classes (Patient, Diagnosis, Treatment, Gene Expression, Mutation)
- Built a Python semantic lifting pipeline transforming the METABRIC dataset (1,904 patients × 693 variables) into an RDF knowledge graph of ~3.85M triples
- Benchmarked AllegroGraph, Blazegraph, and Virtuoso in isolated Docker containers on query latency, RAM, and disk.

Audio Tag Editor | Flutter, BLoC, Android

Aug 2025

- Built and launched a **production-ready Flutter mobile application** for audio metadata editing
- Integrated a custom Flutter plugin with native Android (Java) code for efficient file processing
- Implemented **BLoC architecture** for scalable and maintainable state management
- Achieved **5,000+ installs** and **50+ paying users**, generating real revenue on Google Play

flutter_audio_tagger (Open Source) | Dart, Java, Method Channels 

Jul 2025

- Developed and maintained an **open-source Flutter plugin** for reading and editing audio metadata
- Implemented **platform-specific Android code (Java)** and bridged it using method channels
- Supported multiple formats: MP3, M4A, FLAC, OGG, AIFF, WAV, WMA
- Designed a clean and extensible **Tag data model** for managing metadata (title, artist, album, lyrics, genre, artwork)

CourierFlow | Flutter, Spring Boot, GraphHopper, JWT 

May 2025

- Developed a **Flutter mobile application** for delivery management and route tracking
- Integrated with a **Spring Boot backend API** implementing GraphHopper for route optimization
- Implemented **JWT-based authentication** for secure communication between client and backend
- Enabled efficient route planning and delivery workflow management

CERTIFICATES

- Oracle Certified Professional: Java SE 17 Developer 

SKILLS

Languages — Java, Dart, C, PL/SQL, **Backend** — Spring Boot, REST APIs, JWT, **Data Engineering & Semantic Web** — SPARQL, RDF, OWL, Knowledge Graphs, AllegroGraph, Blazegraph, Virtuoso, **Cloud & DevOps** — Docker, Linux, Git, GitHub, **Mobile** — Flutter, **Core** — Data Structures, Algorithms, Object-Oriented Design

EDUCATION

National School Of Applied Sciences, *Data science and cloud computing*

Les Classes Préparatoires aux Grandes Écoles (CPGE), *Technology and industrial science*